

Linear Regression — Quiz 1 Practice Problems

Solve the following problems. **Show your work.**

1. *Understanding the simple linear regression model.*

For the following questions, indicate whether the statement is true or false, then **explain your answer**. The explanation is worth the majority of the points. You may use examples to support your answer. **You must explain your answer to get full credit.**

- (a) In the simple linear regression model, β_0 and β_1 can be calculated from the data $\{(x_i, y_i)\}_{i=1}^n$.
- (b) The least squares regression line will always go through the center of the data at the point $(\bar{x}, \bar{y})$.
- (c) In simple linear regression, the confidence interval for the mean response at the point $x_0 = \bar{x} - 5$ will be wider than the width of a confidence interval for the mean response at the point $\tilde{x}_0 = \bar{x} - 10$.

2. *Data Analysis and Interpretation.*

Behavioral scientists are interested in how much eating habits affects a bunny's playfulness. They collect a random sample of $n = 20$ rabbits. For each bunny they: 1. feed it a carrot of a specified weight (x , in grams) in line with its usual eating habits, and then 2. measure its hop rate (y , in hops per minute) when released into a grassy area that they can explore.

Two competing ideas exist among the scientists: One idea is that larger carrots may make bunnies have more energy and so they will hop faster. Another idea is that larger carrots may make them full, slowing them down while they digest. We will try to decide which theory is better evidenced by the real data.

A linear model is fit using the carrot weight as a predictor of hop rate. You may assume the classical simple linear regression model assumptions hold.

You are told the following summary statistics:

$$\bar{x} = 10, \quad \bar{y} = 30$$

$$\sum_{i=1}^n (x_i - \bar{x})^2 = 200, \quad \sum_{i=1}^n (y_i - \bar{y})^2 = 1700,$$

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = -400, \quad \text{and} \quad \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 900$$

You can use the following facts: $t_{18,0.95}^* = 1.73$, and $t_{18,0.975}^* = 2.10$ where the notation $t_{df,\gamma}^*$ represents the $100\gamma^{th}$ quantile of the t_{df} distribution, and $F_{1,18,0.95}^* = 4.41$, $F_{18,1,0.95}^* = 247.32$, $F_{1,18,0.975}^* = 6.00$, $F_{18,1,0.975}^* = 990.35$, where the notation $F_{df_1,df_2,\gamma}^*$ represents the $100\gamma^{th}$ quantile of the F_{df_1,df_2} distribution.

- (a) Calculate the least squares estimates for the intercept and slope. Interpret each in the real life context of the problem.
- (b) Calculate the coefficient of determination. Sketch an accompanying scatterplot of what you think the data may look like.
- (c) Perform a hypothesis test which aims to evaluate whether a linear association exists between carrot size and hop rate. State the null and alternative hypotheses, your test statistic, the critical value, and the decision of the test in terms of the real world context of the problem, if the significance level is set to $\alpha = 0.05$. Which of the scientists' ideas were supported by the data?
- (d) Carry out an F-test to assess the hypotheses in part c. State your test statistic, the critical value, and the decision of the test.
- (e) Do your tests in c. and d. agree or disagree? Explain why this is expected or unexpected.

3. Rigorous Derivations.

For the simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where ϵ_i are iid with mean 0 and variance σ^2 . Applying the method of least squares, we obtained the least square estimates $\hat{\beta}_0$ and $\hat{\beta}_1$. Then, the fitted values are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the residuals are $e_i = y_i - \hat{y}_i$.

You can directly take the closed form of the LS estimators as facts:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad \text{and} \quad \hat{\beta}_1 = \frac{SSXY}{SSX}, \quad \text{where} \quad SSXY = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad \text{and} \quad SSX = \sum_{j=1}^n (x_j - \bar{x})^2 \neq 0.$$

However, **any other claims you use must be proven!**

Prove the following, *explaining your reasoning clearly* along the way:

- (a) $\hat{\beta}_0$ is unbiased for β_0 . (Hint: if needed, you can use the fact that $\sum_{i=1}^n (x_i - \bar{x})^2 = (\sum_{i=1}^n x_i^2) - n\bar{x}^2$ freely.)
- (b) $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{SSX}$.
- (c) Bonus. (+10pts) Using the fact that $\text{Var}(\hat{y}_i) = \sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX} \right)$ and any of your results from parts (a) and (b), prove that

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0 \quad \text{if and only if} \quad \bar{x} = 0.$$

Formula Sheet

This is the formula sheet that will be provided with the exam.

SLR Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

where $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ under classical assumptions.

SLR LS Estimators:

$$\hat{\beta}_1 = \frac{SSXY}{SSX} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{j=1}^n (x_j - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Regression Quantities and Relations

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$e = y_i - \hat{y}_i$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$MSE = \frac{\sum_{i=1}^n e_i^2}{n - p}$$

$$R^2 = 1 - SSE/SST$$

Confidence Intervals

General Form:

$$\text{point est.} \pm \text{crit. value} \times \widehat{SE}(\text{point est.})$$

Specific Forms:

$$\hat{\beta}_j \pm \text{crit. value} \times \widehat{SE}(\hat{\beta}_j)$$

$$\hat{y}_h \pm \text{crit. value} \times \widehat{SE}(\hat{y}_h)$$

Prediction Intervals

$$\hat{y}_{new}(x) \pm \text{crit. value} \times \widehat{SE}(\hat{y}_{new}(x) - y_{new}(x))$$

Calculus

$$1. \frac{\partial(f(x)^n)}{\partial x} = n f(x)^{n-1} \frac{\partial f}{\partial x}$$

$$2. \frac{\partial(\log(f(x)))}{\partial x} = \frac{1}{f(x)} \frac{\partial f}{\partial x}$$

$$3. \frac{\partial c}{\partial x} = 0 \text{ where } c \text{ is a constant in } x$$

Distribution Definitions

A random variable Y is said to have the multivariate Normal distribution with mean μ and variance σ^2 if it obeys the following pdf:

$$f_Y(y) = (\sigma\sqrt{2\pi})^{-1} \exp\{-(y - \mu)^2/(2\sigma^2)\}.$$

If a random vector $Z \sim N(0, I_n)$, then the random variable $V := \|Z\|^2 = \sum_{i=1}^n Z_i^2$ follows a chi-squared distribution with n degrees of freedom:

$$V \sim \chi_n^2$$

If U and V are independent random variables and $U \sim \chi_{df_1}^2$ and $V \sim \chi_{df_2}^2$, then the random variable F follows the F_{df_1, df_2} where F is defined as:

$$F = \frac{U/df_1}{V/df_2}$$

Random Variables

If Y is a random variable with mean μ and variance σ^2 and A is a constant, then

$$1. E(AY) = A\mu$$

$$2. \text{Var}(AY) = A^2\sigma^2.$$